Exp.no 2: Bias Audit of Pre-Trained Word Embeddings Using the Word Embedding Association Test (WEAT)

Learning Objective:
Develop a program that performs a bias audit using Word Embedding Association Test (WEAT), to identify and quantify gender or racial stereotypes present in Google News Word2Vec embeddings.

Steps:

1. Import necessary Python libraries: gensim, numpy, etc.
2. Load the Google News Word2Vec embeddings. (2GB Dataset, takes time)
3. Define Target and Attribute Word Sets
   a. Target words (X and Y): Groups to compare, e.g.,
      i. Gender: `X = ["man","male","boy"……],`
         `Y = ["woman","female","girl"……]`
   b. Attribute words (A and B): Concepts to test association with, e.g.,
      i. Career vs Family: `A = ["career","business"……],`
         `B = ["family","home"……]`

4. Define Helper Functions – association function, weat score function, effect size function and compute weat score, effect size.
5. Interpret the Results as

a. Effect Size > 0: Target group X is more associated with A (bias towards X-A).
b. Effect Size < 0: Target group Y is more associated with A (bias towards Y-A).
c. Effect Size ≈ 0: No detectable bias.

Optional Extensions

1. Perform multiple WEAT tests for different bias types (**gender, racial, religious**).
2. Visualize effect sizes using bar charts for comparison.

Program:

```
# Install genism

!pip install gensim

import numpy as np

import gensim

import gensim.downloader as api

import matplotlib.pyplot as plt

# Load pretrained Word2Vec model

print("Loading Word2Vec model...")

model = api.load("word2vec-google-news-300")

print("Model Loaded!")

# Helper Functions
```

```
def filter_vocab(words):
    return [w for w in words if w in model.key_to_index]
def association(w, A, B):
    mean_A = np.mean([model.similarity(w, a) for a in A])
    mean_B = np.mean([model.similarity(w, b) for b in B])
    return mean_A - mean_B
def weat_score(X, Y, A, B):
    score_X = np.sum([association(x, A, B) for x in X])
    score_Y = np.sum([association(y, A, B) for y in Y])
    return score_X - score_Y
def effect_size(X, Y, A, B):
    assoc_X = np.array([association(x, A, B) for x in X])
    assoc_Y = np.array([association(y, A, B) for y in Y])
    mean_diff = np.mean(assoc_X) - np.mean(assoc_Y)
    pooled_std = np.std(np.concatenate([assoc_X, assoc_Y]), ddof=1)
    return mean_diff / pooled_std
def run_weat(test_name, X, Y, A, B)
    X, Y = filter_vocab(X), filter_vocab(Y)
    A, B = filter_vocab(A), filter_vocab(B)
    weat = weat_score(X, Y, A, B)
    eff_size = effect_size(X, Y, A, B)
    print("\n==========", test_name, "==========")
    print("WEAT Score:", weat)
    print("Effect Size:", eff_size)
    if eff_size > 0:
        print("Interpretation: X is more associated with A.")
    elif eff_size < 0:
```

```
        print("Interpretation: Y is more associated with A.")
    else:
        print("Interpretation: No detectable bias.")
    return eff_size
# Word Sets
X_gender = ["man", "male", "boy", "brother", "he", "him", "his", "son"]
Y_gender = ["woman", "female", "girl", "sister", "she", "her", "hers", "daughter"]
A_career = ["career", "business", "office", "salary", "corporation",
            "professional", "management", "industry"]
B_family = ["family", "home", "children", "parent",
            "marriage", "wedding", "relatives", "house"]
X_race = ["european", "caucasian", "white"]
Y_race = ["african", "black"]
A_pleasant = ["joy", "love", "peace", "wonderful",
              "pleasure", "friend", "laughter"]
B_unpleasant = ["pain", "hate", "anger",
                "evil", "failure", "nasty", "awful"]
X_religion = ["christian", "church", "bible"]
Y_religion = ["muslim", "islam", "quran"]
# Run Tests
results = {}
results["Gender"] = run_weat(
    "Gender Bias (Career vs Family)",
    X_gender, Y_gender,
    A_career, B_family
)
results["Racial"] = run_weat(
```

```
    "Racial Bias (Pleasant vs Unpleasant)",
    X_race, Y_race,
    A_pleasant, B_unpleasant
)
results["Religious"] = run_weat(
    "Religious Bias (Pleasant vs Unpleasant)",
    X_religion, Y_religion,
    A_pleasant, B_unpleasant
)
# Plot Results
plt.figure(figsize=(8,5))
plt.bar(results.keys(), results.values())
plt.axhline(0)
plt.title("WEAT Effect Size Comparison")
plt.ylabel("Effect Size")
plt.xlabel("Bias Type")
plt.show()
```

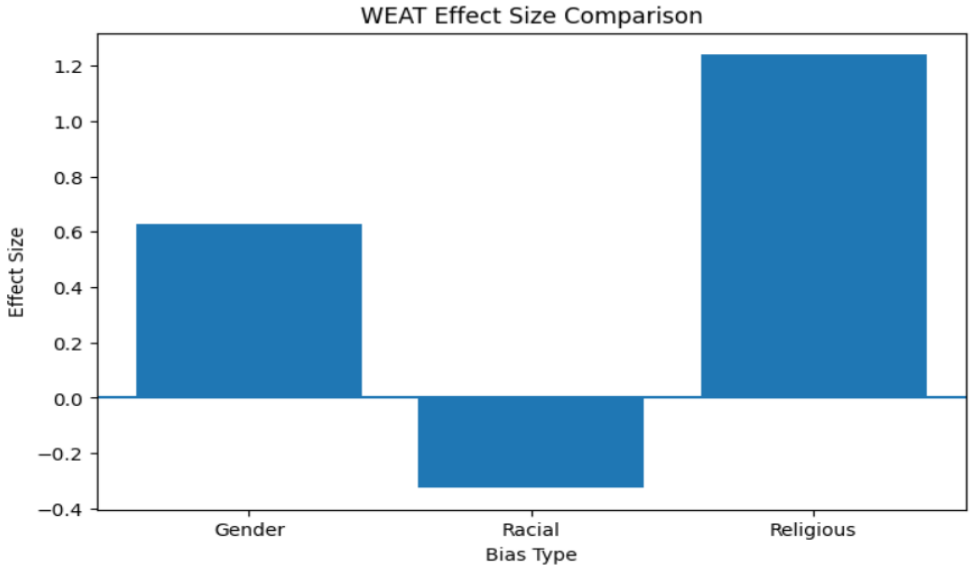
Output :

```
Requirement already satisfied: gensim in /usr/local/lib/python3.12/dist-packages (4.4.0)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.5.1)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim) (2.1.1)
Loading Word2Vec model...
Model Loaded!

========== Gender Bias (Career vs Family) ==========
WEAT Score: 0.5760747
Effect Size: 0.6290184
Interpretation: X is more associated with A.

========== Racial Bias (Pleasant vs Unpleasant) ==========
WEAT Score: -0.07668306
Effect Size: -0.32590735
Interpretation: Y is more associated with A.

========== Religious Bias (Pleasant vs Unpleasant) ==========
WEAT Score: 0.16131353
Effect Size: 1.240433
Interpretation: X is more associated with A.
```



Learning Outcome:

Students will be able to process word embedding datasets, WEAT was implemented to measure associations between target and attribute word sets, interpreted effect sizes to detect bias, and critically evaluated the presence of gender or racial stereotypes in pre-trained embeddings.